# Supporting Information

*Coverage versus Selectivity in Pre-Export Row Scaling for Generated Mixed-Integer Programs*

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This document contains (i) the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text; Figures S1–S3), which mirror the corresponding 1%-MIPGap figures of the main text; (ii) the 0.1%-MIPGap CPLEX distributional and profile views (Scenario 4; Figure S4); (iii) the hyperparameter sensitivity of the structure-aware rule (Table S1); (iv) the per-variant original-scale reliability breakdown with PAR10 under the three readings of the main text (Table S2); (v) the out-of-sample sensitivity of the collective coupling reliability floor (Table S3); (vi) the evaluation of the optional local reliability floor on the failing Simple subset (Table S4); (vii) the common-pool exclusions and full-pool sensitivity (Table S5); (viii) paired-test attrition, the corrected Wilcoxon procedure, and clustered-bootstrap accounting (Table S6); (ix) the available solver-internal-scaling sweep and the verified CPLEX `Read::Scale= −1` audit (Table S7); and (x) the archived runtime comparison used for the coupling-stage activation audit (Table S8); (xi) the direct SA–Flat strict/mixed reliability audit (Table S9); and (xii) the multiplicity, operational-family, bootstrap, preprocessing-plus-solver, deterministic-effort, and own-incumbent sensitivities for that direct comparison, together with its per-contrast attrition and preprocessing-time distributions (Table S10).

Here, PAR10 (including its ITT/strict/mixed superscripts) denotes the common-pool metric defined in the main text. The distinct symbol $\mathrm{PAR10}_{\mathrm{avail}}$ is reserved for archived descriptive revision tables: its denominator is the variant's available solver-output records, a timeout receives 36,000 s, and a pure pipeline error with no solver output is excluded. It must not be compared numerically with the fixed-pool metric.

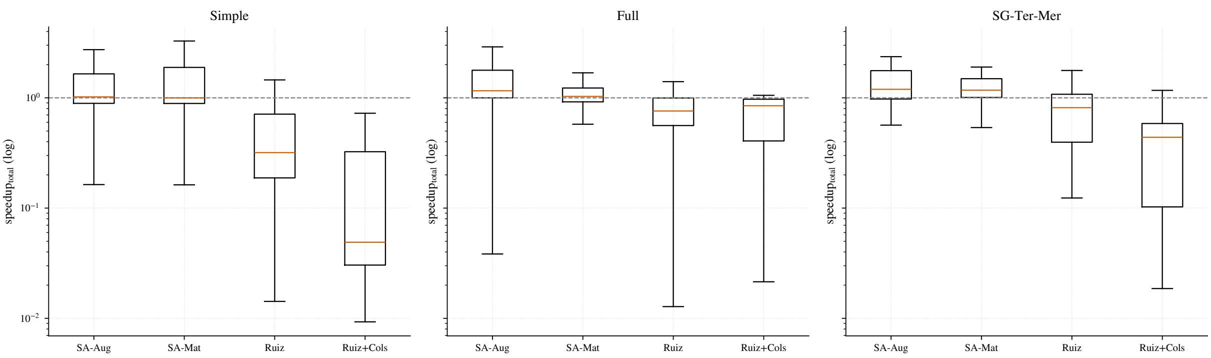
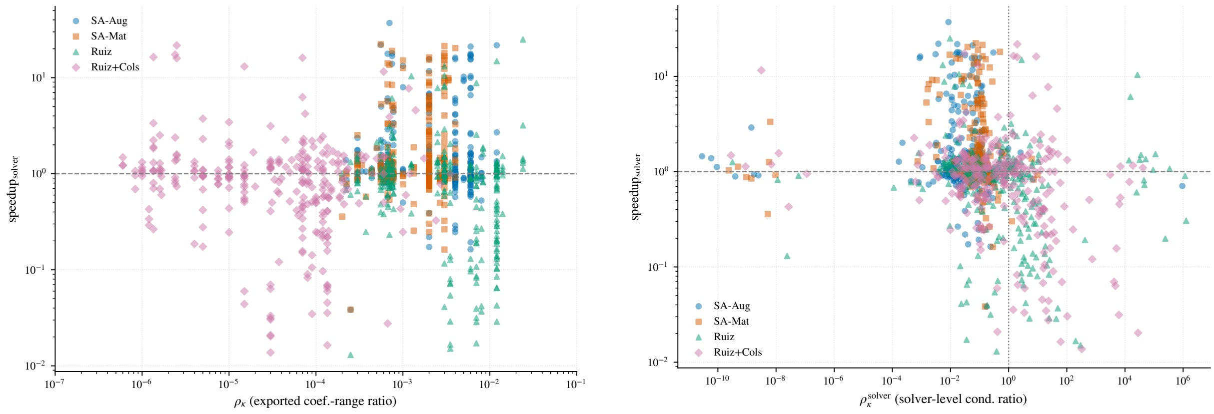



**Figure S1:** At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.




**(a)** Static coefficient-range ratio. **(b)** Solver-level conditioning ratio.

**Figure S2:** At 0.1% MIPGap, the pooled scatter shows a stronger association with speedup for the own-incumbent path-specific fixed-LP diagnostic than for the exported-range proxy. This describes solver paths; it does not make the former a purer measure of representation or an instance-level predictor for the structure-aware variants.

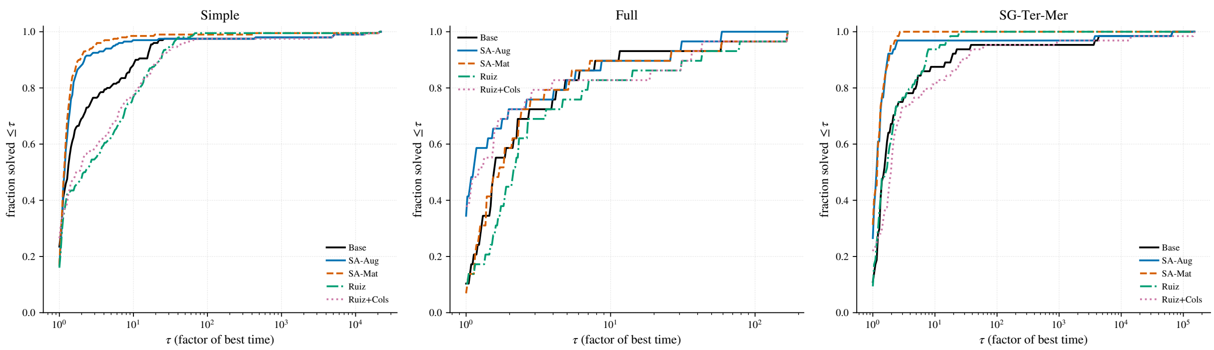
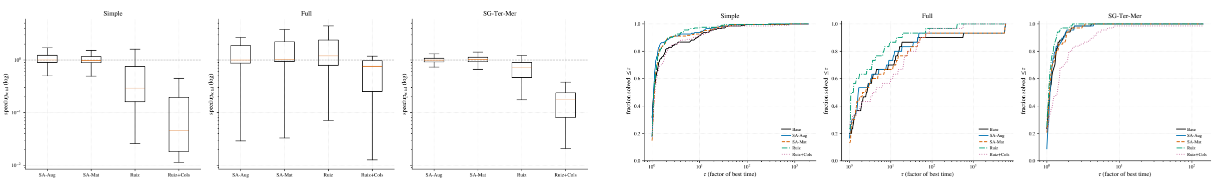



**Figure S3:** At 0.1% MIPGap, fixed-pool Dolan–Moré profiles use the common denominators and individual 36,000-s failure cost defined in the main text.




**(a)** Total-time speedup distribution. **(b)** Fixed-pool performance profile.

**Figure S4:** At 0.1% MIPGap, the CPLEX visual evidence remains mixed under the stricter tolerance. The profile uses a common pool and assigns 36,000 s to an individual timeout, abort, missing run, or invalid time, as defined in the main text.

**Table S1:** Hyperparameter sensitivity of SA-Aug (Simple class, Gurobi, 1% MIPGap). Solver-time sgm ($s = 1$ s) and descriptive $\mathrm{PAR10}_{\mathrm{avail}}$, in seconds; default settings in bold.

| Parameter | Value | sgm (s) | $\mathrm{PAR10}_{\mathrm{avail}}$ (s) |
|---|---|---|---|
| | 1.5 | 2.09 | 2.46 |
| Band $u$ | **2** | 2.07 | 2.45 |
| | 4 | 2.12 | 2.49 |
| | $[10^{-6}, 10^{6}]$ | 2.04 | 2.40 |
| Window $[\varepsilon_{\min}, \varepsilon_{\max}]$ | $[\mathbf{10^{-9}}, \mathbf{10^{9}}]$ | 2.02 | 2.39 |
| | $[10^{-12}, 10^{12}]$ | 2.01 | 2.36 |

Base sgm is 3.73 s for reference. Each row is a separate matched run; the spread across all settings (sgm $\in [2.01, 2.12]$) is far below the base-to-scaled gap, and the median paired speedup is $\approx 1.0$ throughout. The number of Ruiz iterations $K$ is a parameter of the Ruiz baselines, not of the structure-aware rule, so it is fixed at $K = 4$; varying it leaves SA-Aug unchanged, as expected.

**Table S2:** Per-variant audit and PAR10 under the three operational readings. All campaigns use the manifest-declared $T_{\max} = 3600\,\mathrm{s}$. A timeout or pipeline abort receives $10T_{\max} = 36000\,\mathrm{s}$ in every reading; a returned output that fails the relevant verifier receives that penalty only in the strict or mixed reading. The count columns use the raw manifest pool: $n_{\mathrm{att}}$ is every attempted instance and $n_{\mathrm{out}}$ reproduces the pooled diagnostic audit's status-OK denominator, including time-limited runs and three records that lack verification fields. TO is a subset of $n_{\mathrm{out}}$, and Abort is $n_{\mathrm{att}} - n_{\mathrm{out}}$. Pass counts are among those status-OK records; missing verification is non-passing. Strict pass means $v_{\mathrm{row}} \le 10^{-4}$ and $v_{\mathrm{int}} \le 10^{-5}$; mixed pass rescues only row failures with $v_r/(1 + |b_r|) \le 10^{-6}$. In contrast, the four PAR10 columns use the reproducible scenario-specific common pool after the uniformly-unavailable exclusion in Table S5: Base ITT and, for the listed variant, ITT/strict/mixed. Values are seconds, rounded to the nearest second.

| Solver | Gap | Class | Variant | $n_{\mathrm{att}}$ | $n_{\mathrm{out}}$ | TO | Abort | $n^{\mathrm{str}}_{\mathrm{pass}}$ | $n^{\mathrm{mix}}_{\mathrm{pass}}$ | Base ITT | ITT | Strict | Mixed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gurobi | 1% | Simple | SA-Aug | 204 | 197 | 0 | 7 | 185 | 185 | 373 | 364 | 2535 | 2535 |
| Gurobi | 1% | Simple | SA-Mat | 204 | 198 | 0 | 6 | 179 | 180 | 373 | 183 | 3620 | 3439 |
| Gurobi | 1% | Simple | Ruiz | 204 | 198 | 0 | 6 | 143 | 146 | 373 | 207 | 10153 | 9611 |
| Gurobi | 1% | Simple | Ruiz+Cols | 204 | 199 | 0 | 5 | 180 | 189 | 373 | 28 | 3462 | 1835 |
| Gurobi | 1% | SG-Ter-Mer | SA-Aug | 68 | 67 | 1 | 1 | 66 | 67 | 1094 | 1083 | 1613 | 1083 |
| Gurobi | 1% | SG-Ter-Mer | SA-Mat | 68 | 67 | 0 | 1 | 65 | 66 | 1094 | 1079 | 1608 | 1079 |
| Gurobi | 1% | SG-Ter-Mer | Ruiz | 68 | 68 | 1 | 0 | 67 | 68 | 1094 | 567 | 1095 | 567 |
| Gurobi | 1% | SG-Ter-Mer | Ruiz+Cols | 68 | 68 | 1 | 0 | 67 | 68 | 1094 | 585 | 1113 | 585 |
| Gurobi | 1% | Full | SA-Aug | 34 | 34 | 3 | 0 | 27 | 27 | 3385 | 3308 | 9640 | 9640 |
| Gurobi | 1% | Full | SA-Mat | 34 | 34 | 2 | 0 | 31 | 31 | 3385 | 3473 | 5586 | 5586 |
| Gurobi | 1% | Full | Ruiz | 34 | 33 | 4 | 1 | 31 | 31 | 3385 | 5491 | 7577 | 7577 |
| Gurobi | 1% | Full | Ruiz+Cols | 34 | 33 | 4 | 1 | 31 | 32 | 3385 | 5617 | 6666 | 5617 |
| Gurobi | 0.1% | Simple | SA-Aug | 204 | 195 | 0 | 9 | 174 | 178 | 928 | 928 | 4705 | 3987 |
| Gurobi | 0.1% | Simple | SA-Mat | 204 | 198 | 0 | 6 | 168 | 172 | 928 | 381 | 5776 | 5057 |
| Gurobi | 0.1% | Simple | Ruiz | 204 | 199 | 0 | 5 | 155 | 159 | 928 | 221 | 8138 | 7419 |
| Gurobi | 0.1% | Simple | Ruiz+Cols | 204 | 195 | 0 | 9 | 175 | 185 | 928 | 948 | 4544 | 2746 |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Aug | 68 | 63 | 1 | 5 | 62 | 63 | 2246 | 1683 | 2237 | 1683 |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Mat | 68 | 65 | 0 | 3 | 63 | 64 | 2246 | 576 | 1130 | 576 |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz | 68 | 65 | 1 | 3 | 65 | 65 | 2246 | 587 | 587 | 587 |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz+Cols | 68 | 62 | 1 | 6 | 61 | 62 | 2246 | 2262 | 2814 | 2262 |
| Gurobi | 0.1% | Full | SA-Aug | 34 | 30 | 3 | 4 | 27 | 27 | 5992 | 5897 | 8140 | 8140 |
| Gurobi | 0.1% | Full | SA-Mat | 34 | 31 | 4 | 3 | 28 | 30 | 5992 | 6026 | 9293 | 7148 |
| Gurobi | 0.1% | Full | Ruiz | 34 | 30 | 4 | 4 | 27 | 27 | 5992 | 7066 | 9277 | 9277 |
| Gurobi | 0.1% | Full | Ruiz+Cols | 34 | 31 | 5 | 3 | 28 | 29 | 5992 | 7073 | 9312 | 8191 |
| CPLEX | 1% | Simple | SA-Aug | 204 | 204 | 0 | 0 | 37 | 37 | 6 | 3 | 29471 | 29471 |
| CPLEX | 1% | Simple | SA-Mat | 204 | 204 | 0 | 0 | 43 | 43 | 6 | 3 | 28412 | 28412 |
| CPLEX | 1% | Simple | Ruiz | 204 | 204 | 0 | 0 | 44 | 44 | 6 | 4 | 28237 | 28237 |
| CPLEX | 1% | Simple | Ruiz+Cols | 204 | 204 | 0 | 0 | 137 | 138 | 6 | 3 | 11825 | 11649 |
| CPLEX | 1% | SG-Ter-Mer | SA-Aug | 68 | 68 | 0 | 0 | 68 | 68 | 3 | 10 | 10 | 10 |
| CPLEX | 1% | SG-Ter-Mer | SA-Mat | 68 | 68 | 0 | 0 | 68 | 68 | 3 | 6 | 6 | 6 |
| CPLEX | 1% | SG-Ter-Mer | Ruiz | 68 | 68 | 0 | 0 | 68 | 68 | 3 | 10 | 10 | 10 |
| CPLEX | 1% | SG-Ter-Mer | Ruiz+Cols | 68 | 68 | 0 | 0 | 68 | 68 | 3 | 15 | 15 | 15 |

Continued on next page

Table S2 (continued)

| Solver | Gap | Class | Variant | $n_{\text{att}}$ | $n_{\text{out}}$ | TO | Abort | $n_{\text{pass}}^{\text{str}}$ | $n_{\text{pass}}^{\text{mix}}$ | Base ITT | ITT | Strict | Mixed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CPLEX | 1% | Full | SA-Aug | 34 | 34 | 5 | 0 | 25 | 25 | 6515 | 5546 | 15014 | 15014 |
| CPLEX | 1% | Full | SA-Mat | 34 | 34 | 6 | 0 | 25 | 26 | 6515 | 6498 | 14932 | 13875 |
| CPLEX | 1% | Full | Ruiz | 34 | 34 | 3 | 0 | 28 | 28 | 6515 | 3506 | 8744 | 8744 |
| CPLEX | 1% | Full | Ruiz+Cols | 34 | 34 | 4 | 0 | 32 | 33 | 6515 | 4681 | 6793 | 5737 |
| CPLEX | 0.1% | Simple | SA-Aug | 204 | 204 | 1 | 0 | 40 | 40 | 214 | 188 | 29119 | 29119 |
| CPLEX | 0.1% | Simple | SA-Mat | 204 | 204 | 1 | 0 | 38 | 38 | 214 | 203 | 29311 | 29311 |
| CPLEX | 0.1% | Simple | Ruiz | 204 | 204 | 1 | 0 | 44 | 44 | 214 | 192 | 28413 | 28413 |
| CPLEX | 0.1% | Simple | Ruiz+Cols | 204 | 204 | 3 | 0 | 144 | 146 | 214 | 547 | 10955 | 10779 |
| CPLEX | 0.1% | SG-Ter-Mer | SA-Aug | 68 | 68 | 0 | 0 | 68 | 68 | 19 | 21 | 21 | 21 |
| CPLEX | 0.1% | SG-Ter-Mer | SA-Mat | 68 | 68 | 0 | 0 | 68 | 68 | 19 | 16 | 16 | 16 |
| CPLEX | 0.1% | SG-Ter-Mer | Ruiz | 68 | 68 | 0 | 0 | 68 | 68 | 19 | 8 | 8 | 8 |
| CPLEX | 0.1% | SG-Ter-Mer | Ruiz+Cols | 68 | 68 | 1 | 0 | 68 | 68 | 19 | 539 | 539 | 539 |
| CPLEX | 0.1% | Full | SA-Aug | 34 | 34 | 8 | 0 | 24 | 24 | 9951 | 8930 | 18283 | 18283 |
| CPLEX | 0.1% | Full | SA-Mat | 34 | 33 | 10 | 1 | 25 | 26 | 9951 | 11916 | 20280 | 19226 |
| CPLEX | 0.1% | Full | Ruiz | 34 | 34 | 5 | 0 | 32 | 32 | 9951 | 5857 | 7917 | 7917 |
| CPLEX | 0.1% | Full | Ruiz+Cols | 34 | 33 | 10 | 1 | 31 | 32 | 9951 | 11950 | 14063 | 13007 |

*Objective-difference audit.* MIPGap bounds each incumbent against its own dual bound and is not a cross-run agreement test. Of 4,806 variant runs with returned solver output, 72 differ from the corresponding Base incumbent by more than twice the nominal gap; these cases are concentrated in a small set of instance identifiers. The largest pooled relative deviations occur in Full/Ruiz cells ($|\Delta z_{\mathrm{rel}}| \approx 1.8 \times 10^2$ under Gurobi and $5.3 \times 10^1$ under CPLEX). Base is not presumed correct: one Simple Base incumbent is 73% worse than the best verified incumbent, and one CPLEX Full Base incumbent is roughly six times worse. Accordingly, correctness is assessed against the best original-scale-verified incumbent on each instance, with Base subjected to the same feasibility and integrality checks as every scaled variant.

**Table S3:** Out-of-sample sensitivity of the reliability floor for the collective coupling factor (SA-Aug), on twenty seeds (21–40) never used while designing the floor. Cells: coupling, coupling-uniform, and mixed patterns at severity $S = 6$ (60 instances, Gurobi, MIPGap $10^{-3}$, 600 s limit). Median post-scaling global coefficient-and-right-hand-side magnitude-range proxy per pattern; number of runs failing original-scale verification; worst original-scale row violation; instances whose objective disagrees with the base beyond the MIPGap. The failure mode replicates out-of-sample with the floor disabled, and every floor value across three orders of magnitude eliminates it.

| Floor | coupling | coupling-unif. | mixed | $n_{\mathrm{fail}}$ | $v_{\max}$ | over gap |
|---|---|---|---|---|---|---|
| (pre-scaling) | $4.4 \times 10^8$ | $3.3 \times 10^6$ | $1.5 \times 10^{14}$ | | | |
| disabled | $3.8 \times 10^7$ | $5.4 \times 10^3$ | $2.9 \times 10^7$ | 4 | $4.2 \times 10^{-1}$ | 4 |
| $10^{-2}$ | $9.1 \times 10^5$ | $5.4 \times 10^3$ | $2.9 \times 10^6$ | 0 | $1.2 \times 10^{-7}$ | 0 |
| $10^{-3}$ (default) | $3.5 \times 10^5$ | $5.4 \times 10^3$ | $2.3 \times 10^6$ | 0 | $1.2 \times 10^{-7}$ | 0 |
| $10^{-4}$ | $1.0 \times 10^6$ | $5.4 \times 10^3$ | $2.3 \times 10^6$ | 0 | $1.2 \times 10^{-7}$ | 0 |
| $10^{-5}$ | $2.1 \times 10^6$ | $5.4 \times 10^3$ | $3.0 \times 10^6$ | 0 | $1.2 \times 10^{-7}$ | 0 |

*Coupling-trigger sensitivity.* On the original synthetic grid, sweeping $\widehat{\rho}_{\mathrm{trigger}} \in \{2, 5, 10, 20, 50\}$ leaves every displayed coupling and coupling-uniform cell median unchanged. Approximately one instance in twenty lies in the $[2, 50]$ band, so some individual activation decisions change, but the proxy is effectively bimodal (above $10^2$ on oversized instances and below one on undersized instances). The result therefore supports the stated conclusion about the one-sided design, not a privileged choice of the cutoff 10.

**Table S4:** Optional local reliability floor (SA-Aug-LF), evaluated on the twelve Simple instances (Gurobi, 1% MIPGap) whose SA-Aug runs fail original-scale verification. The floor applies the local factor only if $\alpha_r^{A,b} \geq \varepsilon_{\text{floor}}^{\text{loc}}$, so $d_r \geq \varepsilon_{\text{floor}}^{\text{loc}}$ bounds the conditional original-scale residual by $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$; rows below the floor stay at base scale. $n_{\text{fail}}$ is the number of the twelve runs failing the strict absolute criterion; $\sigma_{\text{gm}}$ is the shifted geometric mean solver time ($s = 1$ s); $\widehat{\kappa}_{\text{range}}^{\text{post}}$ is the median exported coefficient-and-right-hand-side magnitude-range proxy; $v_{\max}$ is the worst original-scale row violation. SA-Mat is shown for reference.

| Variant | $n_{\text{fail}}$ | $\sigma_{\text{gm}}$ (s) | $\widehat{\kappa}_{\text{range}}^{\text{post}}$ | $v_{\max}$ |
|---|---|---|---|---|
| Base | 0 | 4.90 | – | 0 |
| SA-Aug (no floor, default) | 12 | 2.47 | $6.6 \times 10^{9}$ | $3.7 \times 10^{-3}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-1}$ | 0 | 2.64 | $5.5 \times 10^{11}$ | $5.3 \times 10^{-5}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-2}$ | 0 | 2.89 | $5.5 \times 10^{11}$ | $7.4 \times 10^{-5}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-3}$ | 6 | 5.52 | $2.0 \times 10^{14}$ | $4.0 \times 10^{-2}$ |
| SA-Aug-LF, $\varepsilon_{\text{floor}}^{\text{loc}} = 10^{-4}$ | 6 | 2.19 | $2.0 \times 10^{14}$ | $5.3 \times 10^{-4}$ |
| SA-Mat | 5 | 2.57 | $4.5 \times 10^{12}$ | $1.9 \times 10^{-3}$ |

*Notes.* The floor is monotone: $n_{\text{fail}}$ falls from 12 (no floor) to 6 (shallow floors $10^{-3}/10^{-4}$, which protect only the most extreme rows) to 0 ($10^{-2}/10^{-1}$), consistent with the residual bound $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$. At the effective floor $10^{-2}$ the variant eliminates every failure while retaining a solver-time advantage over the base ($\sigma_{\text{gm}}$ 2.89 vs 4.90 s), even though it *raises* the exported magnitude-range proxy (the protected rows are exactly the over-compressed large-right-hand-side rows)—a further illustration that this static proxy is a weak predictor of runtime. The threshold is imposed on the representation-dependent factor $\alpha_r^{A,b}$ and is therefore not invariant to a positive re-emission of the same unscaled row. Here it is defined relative to the generator's canonical export in physical units. A representation-invariant alternative could threshold a norm of the resulting scaled row, but it would not inherit the same $\varepsilon/\varepsilon_{\text{floor}}^{\text{loc}}$ residual bound. The $\sigma_{\text{gm}}$ values on twelve instances are noisy and are not a full-class estimate. This subset was selected by a post-treatment outcome (the SA-Aug failures), so the table is a demonstration on the failing set; the full-class reliability and runtime effect, and the behavior on Full, remain untested.

**Table S5:** Uniformly unavailable operational records and sensitivity to retaining them. Panel A applies the documented reproducible data rule: exclude an instance only when all five attempted variants have nonzero pipeline return codes. Panel B restores those cells to the manifest-declared full pool and assigns each a 36000-s penalty. Triplets are ITT/strict/mixed PAR10 in seconds; only scenarios with exclusions are shown because common- and full-pool results otherwise coincide. The within-reading ordering of Base, SA-Aug, and SA-Mat is unchanged (the rounded common-pool tie in Gurobi Simple at 0.1% is broken by less than one second before rounding in the full pool).

**Panel A: identifiers**

| Solver | Gap | Class | $N/n$ | Uniformly unavailable identifiers |
|---|---|---|---|---|
| Gurobi | 1% | Simple | 204/199 | `caso46f`, `caso47a`–`caso47d` |
| Gurobi | 0.1% | Simple | 204/200 | `caso02c`–`caso02e`, `caso22a` |
| Gurobi | 0.1% | SG-Ter-Mer | 68/65 | `instance020`–`instance022` |
| Gurobi | 0.1% | Full | 34/32 | `instance046`, `instance047` |

**Panel B: common pool versus penalized full pool**

| Gap/class | Variant | Pool | ITT | Strict | Mixed |
|---|---|---|---|---|---|
| 1%, Simple ($n$ = 199, $N$ = 204) | Base | common | 373 | 554 | 554 |
| | | full | 1246 | 1422 | 1422 |
| | SA-Aug | common | 364 | 2535 | 2535 |
| | | full | 1238 | 3355 | 3355 |
| | SA-Mat | common | 183 | 3620 | 3439 |
| | | full | 1061 | 4414 | 4237 |
| | Ruiz | common | 207 | 10153 | 9611 |
| | | full | 1085 | 10786 | 10258 |
| | Ruiz+Cols | common | 28 | 3462 | 1835 |
| | | full | 909 | 4260 | 2673 |
| 0.1%, Simple ($n$ = 200, $N$ = 204) | Base | common | 928 | 1108 | 1108 |
| | | full | 1615 | 1792 | 1792 |
| | SA-Aug | common | 928 | 4705 | 3987 |
| | | full | 1616 | 5318 | 4615 |
| | SA-Mat | common | 381 | 5776 | 5057 |
| | | full | 1079 | 6369 | 5664 |
| | Ruiz | common | 221 | 8138 | 7419 |
| | | full | 923 | 8684 | 7980 |
| | Ruiz+Cols | common | 948 | 4544 | 2746 |
| | | full | 1635 | 5161 | 3398 |
| 0.1%, SG-Ter-Mer ($n$ = 65, $N$ = 68) | Base | common | 2246 | 2246 | 2246 |
| | | full | 3735 | 3735 | 3735 |
| | SA-Aug | common | 1683 | 2237 | 1683 |
| | | full | 3197 | 3727 | 3197 |
| | SA-Mat | common | 576 | 1130 | 576 |
| | | full | 2139 | 2668 | 2139 |
| | Ruiz | common | 587 | 587 | 587 |
| | | full | 2149 | 2149 | 2149 |
| | Ruiz+Cols | common | 2262 | 2814 | 2262 |
| | | full | 3750 | 4279 | 3750 |
| 0.1%, Full ($n$ = 32, $N$ = 34) | Base | common | 5992 | 5992 | 5992 |
| | | full | 7757 | 7757 | 7757 |
| | SA-Aug | common | 5897 | 8140 | 8140 |
| | | full | 7668 | 9779 | 9779 |
| | SA-Mat | common | 6026 | 9293 | 7148 |
| | | full | 7789 | 10864 | 8845 |
| | Ruiz | common | 7066 | 9277 | 9277 |
| | | full | 8768 | 10849 | 10849 |
| | Ruiz+Cols | common | 7073 | 9312 | 8191 |
| | | full | 8775 | 10882 | 9827 |

*Note.* $N$ is the manifest pool and $n$ the common pool. In the archived logs for every excluded identifier, Base, SA-Aug, Ruiz, and Ruiz+Cols stop at a Gurobi WLS `Overage for too long` licensing error and report no configured solver. The SA-Mat logs instead record optimal completions written to a separate `*-matricial` result tree, while their aggregate campaign records retain nonzero return codes. Thus the evidence supports a combination of license-capacity failure and incomplete artifact consolidation; the exclusion itself is defined only by the observable all-variant return-code rule and does not assert that every underlying solve failed.

**Table S6:** Pair attrition and corrected Wilcoxon audit. $n$ is the effective matched-pair count. UM denotes the all-variant uniformly-unavailable exclusion; $D_{\mathrm{TO}}$, $D_{\mathrm{A}}$, and $D_{\mathrm{mix}}$ are exclusions because both Base and the listed variant are, respectively, timeouts, pipeline aborts, or different kinds of noncompletion. $C_1$ is a one-sided noncompletion retained at exactly $T_{\max} = 3600\,\mathrm{s}$. There are no missing-time or exact-zero pairs in any row. A one-sided timeout is right-censored at $T_{\max}$; assigning the same cost to a one-sided algorithmic/pipeline abort is an operational-failure convention rather than a censoring claim. The test is two-sided Wilcoxon signed-rank on $t_{\mathrm{variant}} - t_{\mathrm{Base}}$ (SciPy 1.15.3), with exact zeros discarded, tied absolute differences assigned midranks, and no continuity correction. The exact null distribution is used for at most 50 nonzero untied differences; otherwise the tie-corrected normal approximation is used. The displayed structure-aware Simple/SG rows use the approximation and Full rows use the exact distribution; no zero or tied-absolute-difference pair occurs. $r_{\mathrm{rb}} < 0$ favors the listed variant. $q$ is Benjamini–Hochberg adjusted over the twelve tests in each solver-by-gap family (three classes by four variants).

| Solver | Gap | Class | Variant | $n$ | UM | $D_{\mathrm{TO}}$ | $D_{\mathrm{A}}$ | $D_{\mathrm{mix}}$ | $C_1$ | $W$ | $p$ | $q$ | $r_{\mathrm{rb}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gurobi | 1% | Simple | SA-Aug | 197 | 5 | 0 | 2 | 0 | 0 | 7535 | $5.67 \times 10^{-3}$ | $9.72 \times 10^{-3}$ | −0.227 |
| Gurobi | 1% | Simple | SA-Mat | 198 | 5 | 0 | 1 | 0 | 1 | 3011 | $2.42 \times 10^{-17}$ | $2.90 \times 10^{-16}$ | −0.694 |
| Gurobi | 1% | Simple | Ruiz | 198 | 5 | 0 | 1 | 0 | 1 | 7248 | $1.27 \times 10^{-3}$ | $3.04 \times 10^{-3}$ | +0.264 |
| Gurobi | 1% | Simple | Ruiz+Cols | 199 | 5 | 0 | 0 | 0 | 2 | 7414 | $1.82 \times 10^{-3}$ | $3.65 \times 10^{-3}$ | +0.255 |
| Gurobi | 1% | SG-Ter-Mer | SA-Aug | 66 | 0 | 1 | 1 | 0 | 0 | 501 | $1.13 \times 10^{-4}$ | $4.51 \times 10^{-4}$ | −0.547 |
| Gurobi | 1% | SG-Ter-Mer | SA-Mat | 66 | 0 | 0 | 1 | 1 | 0 | 115 | $2.49 \times 10^{-10}$ | $1.50 \times 10^{-9}$ | −0.896 |
| Gurobi | 1% | SG-Ter-Mer | Ruiz | 67 | 0 | 1 | 0 | 0 | 1 | 1129 | 0.950 | 0.950 | −0.009 |
| Gurobi | 1% | SG-Ter-Mer | Ruiz+Cols | 67 | 0 | 1 | 0 | 0 | 1 | 598 | $7.26 \times 10^{-4}$ | $2.18 \times 10^{-3}$ | +0.475 |
| Gurobi | 1% | Full | SA-Aug | 32 | 0 | 2 | 0 | 0 | 2 | 175 | 0.0984 | 0.148 | −0.337 |
| Gurobi | 1% | Full | SA-Mat | 32 | 0 | 2 | 0 | 0 | 2 | 204 | 0.270 | 0.360 | −0.227 |
| Gurobi | 1% | Full | Ruiz | 31 | 0 | 2 | 0 | 1 | 2 | 197 | 0.327 | 0.393 | +0.206 |
| Gurobi | 1% | Full | Ruiz+Cols | 32 | 0 | 2 | 0 | 0 | 4 | 243 | 0.705 | 0.769 | +0.080 |
| Gurobi | 0.1% | Simple | SA-Aug | 195 | 4 | 0 | 5 | 0 | 0 | 6963 | $1.02 \times 10^{-3}$ | $3.06 \times 10^{-3}$ | −0.271 |
| Gurobi | 0.1% | Simple | SA-Mat | 198 | 4 | 0 | 2 | 0 | 3 | 7069 | $5.70 \times 10^{-4}$ | $2.28 \times 10^{-3}$ | −0.282 |
| Gurobi | 0.1% | Simple | Ruiz | 199 | 4 | 0 | 1 | 0 | 4 | 7429 | $1.94 \times 10^{-3}$ | $4.66 \times 10^{-3}$ | +0.253 |
| Gurobi | 0.1% | Simple | Ruiz+Cols | 199 | 4 | 0 | 1 | 0 | 8 | 7517 | $2.78 \times 10^{-3}$ | $5.46 \times 10^{-3}$ | +0.245 |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Aug | 62 | 3 | 1 | 2 | 0 | 1 | 456 | $2.63 \times 10^{-4}$ | $1.58 \times 10^{-3}$ | −0.533 |
| Gurobi | 0.1% | SG-Ter-Mer | SA-Mat | 64 | 3 | 0 | 0 | 1 | 3 | 341 | $2.95 \times 10^{-6}$ | $3.53 \times 10^{-5}$ | −0.672 |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz | 64 | 3 | 1 | 0 | 0 | 3 | 889 | 0.313 | 0.341 | −0.145 |
| Gurobi | 0.1% | SG-Ter-Mer | Ruiz+Cols | 64 | 3 | 1 | 0 | 0 | 6 | 599 | $3.19 \times 10^{-3}$ | $5.46 \times 10^{-3}$ | +0.424 |
| Gurobi | 0.1% | Full | SA-Aug | 28 | 2 | 3 | 1 | 0 | 2 | 124 | 0.0735 | 0.110 | −0.389 |
| Gurobi | 0.1% | Full | SA-Mat | 27 | 2 | 4 | 1 | 0 | 0 | 173 | 0.714 | 0.714 | −0.085 |
| Gurobi | 0.1% | Full | Ruiz | 28 | 2 | 3 | 0 | 1 | 3 | 144 | 0.186 | 0.248 | +0.291 |
| Gurobi | 0.1% | Full | Ruiz+Cols | 28 | 2 | 4 | 0 | 0 | 3 | 149 | 0.227 | 0.272 | −0.266 |
| CPLEX | 1% | Simple | SA-Aug | 204 | 0 | 0 | 0 | 0 | 0 | 9377 | 0.202 | 0.346 | −0.103 |
| CPLEX | 1% | Simple | SA-Mat | 204 | 0 | 0 | 0 | 0 | 0 | 7460 | $3.89 \times 10^{-4}$ | $2.33 \times 10^{-3}$ | +0.286 |
| CPLEX | 1% | Simple | Ruiz | 204 | 0 | 0 | 0 | 0 | 0 | 9496 | 0.256 | 0.384 | +0.092 |
| CPLEX | 1% | Simple | Ruiz+Cols | 204 | 0 | 0 | 0 | 0 | 0 | 8958 | 0.0762 | 0.183 | +0.143 |
| CPLEX | 1% | SG-Ter-Mer | SA-Aug | 68 | 0 | 0 | 0 | 0 | 0 | 1068 | 0.521 | 0.521 | +0.090 |
| CPLEX | 1% | SG-Ter-Mer | SA-Mat | 68 | 0 | 0 | 0 | 0 | 0 | 740 | $8.15 \times 10^{-3}$ | 0.0326 | +0.369 |

Continued on next page

Table S6 (continued)

| Solver | Gap | Class | Variant | $n$ | UM | $D_{\mathrm{TO}}$ | $D_{\mathrm{A}}$ | $D_{\mathrm{mix}}$ | $C_1$ | $W$ | $p$ | $q$ | $r_{\mathrm{rb}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CPLEX | 1% | SG-Ter-Mer | Ruiz | 68 | 0 | 0 | 0 | 0 | 0 | 1041 | 0.420 | 0.458 | +0.113 |
| CPLEX | 1% | SG-Ter-Mer | Ruiz+Cols | 68 | 0 | 0 | 0 | 0 | 0 | 438 | $7.09 \times 10^{-6}$ | $8.50 \times 10^{-5}$ | +0.627 |
| CPLEX | 1% | Full | SA-Aug | 29 | 0 | 4 | 0 | 1 | 1 | 172 | 0.336 | 0.428 | +0.209 |
| CPLEX | 1% | Full | SA-Mat | 28 | 0 | 5 | 0 | 1 | 0 | 131 | 0.104 | 0.208 | −0.355 |
| CPLEX | 1% | Full | Ruiz | 31 | 0 | 2 | 0 | 1 | 3 | 200 | 0.357 | 0.428 | −0.194 |
| CPLEX | 1% | Full | Ruiz+Cols | 30 | 0 | 3 | 0 | 1 | 2 | 130 | 0.0345 | 0.104 | +0.441 |
| CPLEX | 0.1% | Simple | SA-Aug | 204 | 0 | 0 | 0 | 0 | 2 | 8765 | 0.0453 | 0.121 | −0.162 |
| CPLEX | 0.1% | Simple | SA-Mat | 203 | 0 | 1 | 0 | 0 | 0 | 10056 | 0.723 | 0.723 | −0.029 |
| CPLEX | 0.1% | Simple | Ruiz | 204 | 0 | 0 | 0 | 0 | 2 | 9364 | 0.196 | 0.356 | −0.104 |
| CPLEX | 0.1% | Simple | Ruiz+Cols | 203 | 0 | 1 | 0 | 0 | 2 | 8092 | $6.98 \times 10^{-3}$ | 0.0419 | +0.218 |
| CPLEX | 0.1% | SG-Ter-Mer | SA-Aug | 68 | 0 | 0 | 0 | 0 | 0 | 1085 | 0.591 | 0.709 | +0.075 |
| CPLEX | 0.1% | SG-Ter-Mer | SA-Mat | 68 | 0 | 0 | 0 | 0 | 0 | 985 | 0.251 | 0.372 | −0.160 |
| CPLEX | 0.1% | SG-Ter-Mer | Ruiz | 68 | 0 | 0 | 0 | 0 | 0 | 853 | 0.0505 | 0.121 | −0.273 |
| CPLEX | 0.1% | SG-Ter-Mer | Ruiz+Cols | 68 | 0 | 0 | 0 | 0 | 1 | 473 | $1.89 \times 10^{-5}$ | $2.27 \times 10^{-4}$ | +0.597 |
| CPLEX | 0.1% | Full | SA-Aug | 27 | 0 | 6 | 0 | 1 | 3 | 173 | 0.714 | 0.723 | −0.085 |
| CPLEX | 0.1% | Full | SA-Mat | 26 | 0 | 6 | 0 | 2 | 4 | 125 | 0.208 | 0.356 | −0.288 |
| CPLEX | 0.1% | Full | Ruiz | 30 | 0 | 3 | 0 | 1 | 6 | 134 | 0.0427 | 0.121 | −0.424 |
| CPLEX | 0.1% | Full | Ruiz+Cols | 27 | 0 | 6 | 0 | 1 | 6 | 143 | 0.279 | 0.372 | +0.243 |

*One-sided-abort sensitivity.* Replacing the primary $T_{\max}$ convention by $10T_{\max}$ for a one-sided algorithmic/pipeline abort leaves every effect direction and inferential decision unchanged. The largest change among the displayed Gurobi structure-aware rows is Full/SA-Aug at $0.1\%$: $W = 124 \to 125$, $p = 0.07354 \to 0.07742$, $q_{\mathrm{BH}} = 0.11031 \to 0.11612$, and $r_{\mathrm{rb}} = -0.389 \to -0.384$. All other Gurobi structure-aware rows are unchanged; CPLEX structure-aware $W$, $p$, and $r_{\mathrm{rb}}$ are unchanged, with only minor BH shifts caused by other members of the twelve-test family. The sensitivity is reproduced by `scripts/paired_tests.py --abort-penalty par10`.

*Cluster-bootstrap accounting for the conditioning comparison.* At 1%, 298 of 306 instances have at least one complete conditioning–speedup pair; eight are lost because Base errored. The whole-instance bootstrap uses 1185 complete variant–instance observations (seven of the possible $298 \times 4$ lack the conditioning diagnostic). At $0.1\%$, 288 instances contribute; eighteen Base errors remove nine Simple, six SG-Ter-Mer, and three Full instances. Of 1152 potential rows, 1135 enter the joint comparison: twelve lack a usable variant speedup and five more lack the solver-conditioning diagnostic. Each of the 2000 resamples keeps all available formulation rows of an instance together. At 1%, the within-class solver-level coefficients are $-0.31$ (Simple), $-0.13$ (SG-Ter-Mer), and $-0.12$ (Full); their varying magnitudes reinforce that the pooled coefficient is descriptive rather than an instance-level predictor.

*Exploratory dependence check.* If repeated rows and seeds are naively treated as independent observations, the first-order partial rank correlation between runtime improvement and the solver-facing conditioning diagnostic is approximately $-0.25$ after controlling for loaded nonzeros, and a Williams–Steiger comparison gives $Z \approx 6.1$. Because both calculations ignore the clustered generator design, they are reported only as descriptive diagnostics and support no confirmatory claim.

**Table S7:** Available solver-internal-scaling sweep over both solvers, both MIPGap settings, and all three classes. Each cell is solver-time sgm ($s$ = 1 s)/PAR10$_{\text{avail}}$ in seconds, using the descriptive revision-campaign convention (timeouts receive 36000 s; pure pipeline errors are excluded). SA-Aug and SA-Mat retain the solver default. For Gurobi, Off/A/B/C mean `ScaleFlag`=0/1/2/3. For CPLEX they mean `Read::Scale`=−1/0/1/not applicable. The CPLEX Full 0.1% Scale=0 artifact is partial (26 of 34 rows), and Scale=1 is absent because the four-day sweep budget expired; all other configuration folders contain the manifest-declared number of rows. The lower panel audits the apparently fast CPLEX Scale=−1 Full cell directly from the stored original-scale verifier fields.

**Panel A: timing sweep** (sgm/PAR10$_{\text{avail}}$)

| Solver | Gap | Class | Base | SA-Aug | SA-Mat | Off | A | B | C |
|---|---|---|---|---|---|---|---|---|---|
| Gurobi | 1% | Simple | 3.936/11.26 | 2.133/2.50 | 2.166/2.56 | 41.969/1541.92 | 5.811/20.39 | 4.402/12.60 | 3.791/10.79 |
| Gurobi | 1% | SG-Ter-Mer | 14.087/564.33 | 8.282/554.10 | 8.635/554.64 | 56.625/740.17 | 14.664/566.35 | 20.907/589.78 | 21.475/580.46 |
| Gurobi | 1% | Full | 109.375/3448.17 | 95.239/3349.65 | 102.025/3458.10 | 561.945/9716.08 | 128.449/2444.67 | 138.112/2497.10 | 133.828/3468.97 |
| Gurobi | 0.1% | Simple | 9.656/36.16 | 6.419/210.83 | 6.248/202.17 | 89.358/2031.10 | 15.299/219.91 | 10.607/33.11 | 9.438/35.10 |
| Gurobi | 0.1% | SG-Ter-Mer | 14.587/566.63 | 8.811/554.42 | 9.066/555.19 | 54.784/727.17 | 16.972/620.44 | 11.753/32.85 | 23.102/585.62 |
| Gurobi | 0.1% | Full | 437.243/4766.52 | 388.446/4714.38 | 456.365/4821.25 | 396.978/9577.84 | 337.331/4621.00 | 347.613/3687.08 | 353.389/4697.74 |
| CPLEX | 1% | Simple | 2.621/7.24 | 1.725/3.65 | 1.861/4.01 | 2.500/3.93 | 2.549/7.59 | 2.431/7.18 | – |
| CPLEX | 1% | SG-Ter-Mer | 2.253/3.57 | 2.400/10.26 | 2.309/5.94 | 5.168/12.29 | 2.229/3.53 | 2.175/5.66 | – |
| CPLEX | 1% | Full | 147.821/5609.27 | 127.360/3616.84 | 120.881/6509.40 | 14.381/35.45 | 159.012/5621.84 | 189.129/5644.99 | – |
| CPLEX | 0.1% | Simple | 5.143/215.88 | 4.266/188.91 | 4.450/203.68 | 7.634/209.70 | 5.203/216.59 | 5.625/369.68 | – |
| CPLEX | 0.1% | SG-Ter-Mer | 2.664/22.29 | 2.678/24.19 | 2.667/19.50 | 5.844/13.69 | 2.725/23.13 | 2.477/11.78 | – |
| CPLEX | 0.1% | Full | 425.043/9419.41 | 426.712/9179.82 | 387.204/11235.11 | 13.476/24.97 | 303.376/6196.93† | – | – |

**Panel B: CPLEX Full, `Read::Scale`= −1, original-scale audit**

| Gap | Runs | Strict pass | Mixed pass | max $v_{\text{row}}$ | max $v_{\text{int}}$ | max $\lvert z_{\text{solver}} - z_{\text{recomputed}}\rvert$ | Strict-fail identifiers |
|---|---|---|---|---|---|---|---|
| 1% | 34 | 26 | 26 | $1.3149 \times 10^{-2}$ | 0 | $3.73 \times 10^{-8}$ | `002,003,004,012,014,022,023,044` |
| 0.1% | 34 | 26 | 26 | $1.3149 \times 10^{-2}$ | 0 | $3.73 \times 10^{-8}$ | `002,003,004,012,014,022,023,044` |

*Notes.* The raw feasibility, integrality, and recomputed-objective fields are present for all 34 runs; the two gap folders contain identical objectives and verifier values. At 1%, 11 of 34 Scale=−1 objectives differ from the default-base objective by more than twice the stopping tolerance; six of those 11 also fail strict verification, while five pass it. Thus the speedup is real as timing but cannot be classified as a reliable solution improvement. The campaign's generic `mip_equivalente` field labels every variant named `base` as `BASE`, irrespective of its internal-scaling parameter, so the pass counts above are rebuilt from the raw verifier fields rather than that label. †Partial artifact; its descriptive value is not comparable to a complete 34-instance cell.

**Table S8:** Archived runtime comparison for the operational coupling-stage activation campaign (Gurobi, 1% MIPGap). Solver-time shifted geometric mean (sgm, $s = 1$ s) and descriptive $\mathrm{PAR10}_{\mathrm{avail}}$ are seconds. The accompanying activation diagnostic found zero coupling-stage activations and byte-identical exports for each full/local-only pair; their small runtime differences therefore measure run-to-run variability, not a coupling-stage effect. The coupling-only variant coincides with Base at the exported-model level.

| Class | Variant | sgm (s) | $\mathrm{PAR10}_{\mathrm{avail}}$ (s) |
|---|---|---|---|
| Simple | Base | 3.73 | 10.47 |
| | SA-Aug (local+coupling) | 2.02 | 2.42 |
| | SA-Aug, local only | 2.01 | 2.38 |
| | SA-Aug, coupling only | 3.74 | 10.44 |
| | SA-Mat (local+coupling) | 2.04 | 2.43 |
| | SA-Mat, local only | 2.06 | 2.48 |
| | SA-Mat, coupling only | 3.73 | 10.26 |
| Full | Base | 82.55 | 3379.7 |
| | SA-Aug (local+coupling) | 71.41 | 3315.9 |
| | SA-Aug, local only | 70.94 | 3314.4 |
| | SA-Aug, coupling only | 83.69 | 3381.5 |
| | SA-Mat (local+coupling) | 78.75 | 4384.9 |
| | SA-Mat, local only | 78.88 | 4386.1 |
| | SA-Mat, coupling only | 90.21 | 3394.1 |
| SG-Ter-Mer | Base | 14.42 | 565.6 |
| | SA-Aug (local+coupling) | 8.55 | 555.1 |
| | SA-Aug, local only | 8.58 | 555.2 |
| | SA-Aug, coupling only | 14.45 | 565.9 |
| | SA-Mat (local+coupling) | 8.96 | 555.6 |
| | SA-Mat, local only | 8.83 | 555.6 |
| | SA-Mat, coupling only | 14.43 | 565.8 |

In SG-Ter-Mer one near-limit instance times out under every variant in this independent batch, so $\mathrm{PAR10}_{\mathrm{avail}}$ is dominated by a single 36000-s penalty. In Full, three or four runs time out per variant, so the raw sgm values use different solved subsets and are descriptive. The attribution is the export/activation result: full equals local only, and coupling only equals Base.

**Table S9:** Direct SA–Flat reliability audit under Gurobi. Each row uses all available records from one dedicated class–tolerance batch; both SA and Flat were rerun in that batch. Timeouts (TO), strict/mixed original-scale verification failures (S/M), and fixed-pool PAR10 are reported for both coverage policies. Each PAR10 cell is ITT/strict/mixed in seconds, on the capped solver-time cost $c^{(k,\text{solver})}$ of the main text; recomputing with the preprocessing-plus-solver cost shifts every cell by less than 0.2 s and changes no ordering.

| Gap | Class | Kernel | $N$ | TO SA/Flat | SA fail S/M | Flat fail S/M | SA PAR10 I/S/M | Flat PAR10 I/S/M |
|---|---|---|---|---|---|---|---|---|
| 1% | Simple | Augmented | 204 | 0/0 | 12/12 | 13/12 | 2.4/2119.8/2119.8 | 2.4/2296.2/2119.7 |
| | Simple | Matrix-only | 204 | 0/0 | 19/18 | 19/18 | 2.4/3355.1/3178.7 | 2.4/3355.1/3178.6 |
| | SG-Ter-Mer | Augmented | 68 | 1/1 | 1/0 | 2/0 | 554.5/1083.9/554.5 | 541.1/1598.9/541.1 |
| | SG-Ter-Mer | Matrix-only | 68 | 1/1 | 1/0 | 1/0 | 555.0/1083.7/555.0 | 551.6/1080.8/551.6 |
| | Full | Augmented | 34 | 3/3 | 6/6 | 3/3 | 3312.1/9643.7/9643.7 | 3374.3/6464.9/6464.9 |
| | Full | Matrix-only | 34 | 3/3 | 2/2 | 4/4 | 3420.9/5533.6/5533.6 | 3430.6/7554.0/7554.0 |
| 0.1% | Simple | Augmented | 204 | 1/0 | 22/18 | 25/22 | 211.9/4090.0/3386.8 | 30.3/4436.5/3908.1 |
| | Simple | Matrix-only | 204 | 1/1 | 30/26 | 30/26 | 203.5/5491.9/4787.2 | 203.7/5492.1/4787.4 |
| | SG-Ter-Mer | Augmented | 68 | 1/1 | 1/0 | 1/0 | 556.0/1085.4/556.0 | 542.4/1071.4/542.4 |
| | SG-Ter-Mer | Matrix-only | 68 | 1/1 | 1/0 | 1/0 | 556.5/1085.6/556.5 | 552.8/1082.0/552.8 |
| | Full | Augmented | 30 | 3/3 | 2/2 | 2/1 | 3951.0/6342.2/6342.2 | 4004.3/6290.8/5109.3 |
| | Full | Matrix-only | 30 | 4/4 | 2/1 | 0/0 | 5193.8/7585.0/6389.5 | 5154.8/5154.8/5154.8 |

*Notes.* There are no algorithmic or pipeline aborts in these direct batches. Strict penalizes every strict original-scale failure; mixed penalizes only a failure that also lacks a reliable counterpart on the same instance, as defined in the main text. At Full 0.1%, instances 045–048 are absent from both policies; assigning the same penalty to both missing records changes neither their ordering nor any paired contrast. The key reliability sensitivity is visible in Simple/Augmented: the small Flat solver-time advantage at 0.1% reverses under both strict and mixed costs. SG-Ter-Mer/Augmented retains a small descriptive Flat advantage because neither policy has a mixed failure.

**Table S10:** Magnitude, dependence, and attrition sensitivities for the direct SA–Flat contrasts (exploratory endpoint hierarchy; see the main text). Panel A reports instance-level and operational-family inference for the two central 0.1% augmented contrasts on both endpoints (preprocessing-plus-solver operational endpoint, capped solver-time decomposition), on the shared paired variable $d_i = \log(T_{\mathrm{Flat,i}}/T_{\mathrm{SA,i}})$, with the Hodges–Lehmann pseudomedian expressed as the typical SA/Flat ratio $\exp(-\widehat{\theta}_{\mathrm{HL}})$. Panel B repeats the family analysis for Gurobi Work units. Panel C relocates the path-specific diagnostic from the main timing table because it does not hold the integer solution fixed across policies. Panel D gives the per-contrast attrition and preprocessing-time distributions of the main flat-control table.

**Panel A: paired log time ratio and dependence (central augmented 0.1% contrasts)**

| Class | Endpoint | $n/G$ | med. SA/Flat | HL SA/Flat | family-bootstrap 95% CI | $p_{\mathrm{inst}}$ | $q_6$ | $q_{12}$ | $p_{\mathrm{block}}$ | $q_{6,\mathrm{block}}$ | $q_{12,\mathrm{block}}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Simple | pre+solve | 204/34 | 1.0320 | 1.0359 | [1.0274, 1.0418] | $3.3 \times 10^{-11}$ | $2.0 \times 10^{-10}$ | $1.3 \times 10^{-10}$ | $5.1 \times 10^{-8}$ | $3.1 \times 10^{-7}$ | $2.1 \times 10^{-7}$ |
| Simple | solver | 204/34 | 1.0093 | 1.0135 | [1.0019, 1.0212] | 0.00173 | 0.01038 | 0.02075 | 0.00266 | 0.01595 | 0.03190 |
| SG-Ter-Mer | pre+solve | 67/34 | 1.0473 | 1.1577 | [1.0170, 1.2065] | 0.00031 | 0.00063 | 0.00063 | 0.00056 | 0.00111 | 0.00117 |
| SG-Ter-Mer | solver | 67/34 | 1.0159 | 1.1229 | [0.9797, 1.1309] | 0.02616 | 0.05232 | 0.09346 | 0.03320 | 0.06640 | 0.11980 |

**Panel B: deterministic effort** ($\mathrm{Work}_{\mathrm{Flat}} - \mathrm{Work}_{\mathrm{SA}}$)

| Class | $n$ | Exact zeros | med. Work SA | med. Work Flat | $r_{\mathrm{rb}}$ | $p_{\mathrm{block}}$ | $q_{6,\mathrm{block}}$ | $q_{12,\mathrm{block}}$ |
|---|---|---|---|---|---|---|---|---|
| Simple | 204 | 138 | 2 | 2 | −0.382 | 0.01196 | 0.03587 | 0.04782 |
| SG-Ter-Mer | 67 | 31 | 7 | 4 | −0.649 | 0.000970 | 0.00582 | 0.00637 |

**Panel C: median own-incumbent path-diagnostic ratio**

| Gap | Class | Kernel | $n$ | med. $\rho^{\mathrm{solver}}_{\kappa,\mathrm{SA}}/\rho^{\mathrm{solver}}_{\kappa,\mathrm{Flat}}$ |
|---|---|---|---|---|
| 1% | Simple | Augmented | 204 | 1.000 |
| | Simple | Matrix-only | 204 | 1.000 |
| | SG-Ter-Mer | Augmented | 67 | 0.626 |
| | SG-Ter-Mer | Matrix-only | 67 | 0.849 |
| | Full | Augmented | 31 | 0.628 |
| | Full | Matrix-only | 32 | 0.849 |
| 0.1% | Simple | Augmented | 204 | 1.000 |
| | Simple | Matrix-only | 203 | 1.000 |
| | SG-Ter-Mer | Augmented | 67 | 0.629 |
| | SG-Ter-Mer | Matrix-only | 67 | 0.849 |
| | Full | Augmented | 27 | 0.634 |
| | Full | Matrix-only | 27 | 0.849 |

**Panel D: per-contrast attrition and preprocessing-time distributions**

| Gap | Class | Kernel | Nominal | Recorded | Common pool | Double noncompl. | $n$ used | med. $T^{\mathrm{SA}}_{\mathrm{pre}}$ (s) | med. $T^{\mathrm{Flat}}_{\mathrm{pre}}$ (s) | med. diff (s) |
|---|---|---|---|---|---|---|---|---|---|---|
| 1% | Simple | Augmented | 204 | 204 | 204 | 0 | 204 | 0.074 | 0.025 | +0.050 |
| | Simple | Matrix-only | 204 | 204 | 204 | 0 | 204 | 0.077 | 0.025 | +0.053 |
| | SG-Ter-Mer | Augmented | 68 | 68 | 68 | 1 | 67 | 0.049 | 0.019 | +0.031 |
| | SG-Ter-Mer | Matrix-only | 68 | 68 | 68 | 1 | 67 | 0.051 | 0.019 | +0.033 |
| | Full | Augmented | 34 | 34 | 34 | 3 | 31 | 0.121 | 0.041 | +0.080 |
| | Full | Matrix-only | 34 | 34 | 34 | 2 | 32 | 0.125 | 0.041 | +0.084 |
| 0.1% | Simple | Augmented | 204 | 204 | 204 | 0 | 204 | 0.077 | 0.025 | +0.051 |
| | Simple | Matrix-only | 204 | 204 | 204 | 1 | 203 | 0.079 | 0.025 | +0.054 |
| | SG-Ter-Mer | Augmented | 68 | 68 | 68 | 1 | 67 | 0.051 | 0.019 | +0.032 |
| | SG-Ter-Mer | Matrix-only | 68 | 68 | 68 | 1 | 67 | 0.052 | 0.019 | +0.033 |
| | Full | Augmented | 34 | 30 | 30 | 3 | 27 | 0.120 | 0.041 | +0.079 |
| | Full | Matrix-only | 34 | 30 | 30 | 3 | 27 | 0.124 | 0.041 | +0.083 |